

Prolab II — Proje Sunum Gorselleri

KNN vs Karar Agaci Karsilastirmali Analizi

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Icerik

1. Genel Mimari

— 6 paket, 12 sinif

2. Veri Akis Hatti

— CSV'den sonuca

3. OOP Prensipleri

— 4 prensip

4. KNN Algoritmasi

— Mesafe + oylama

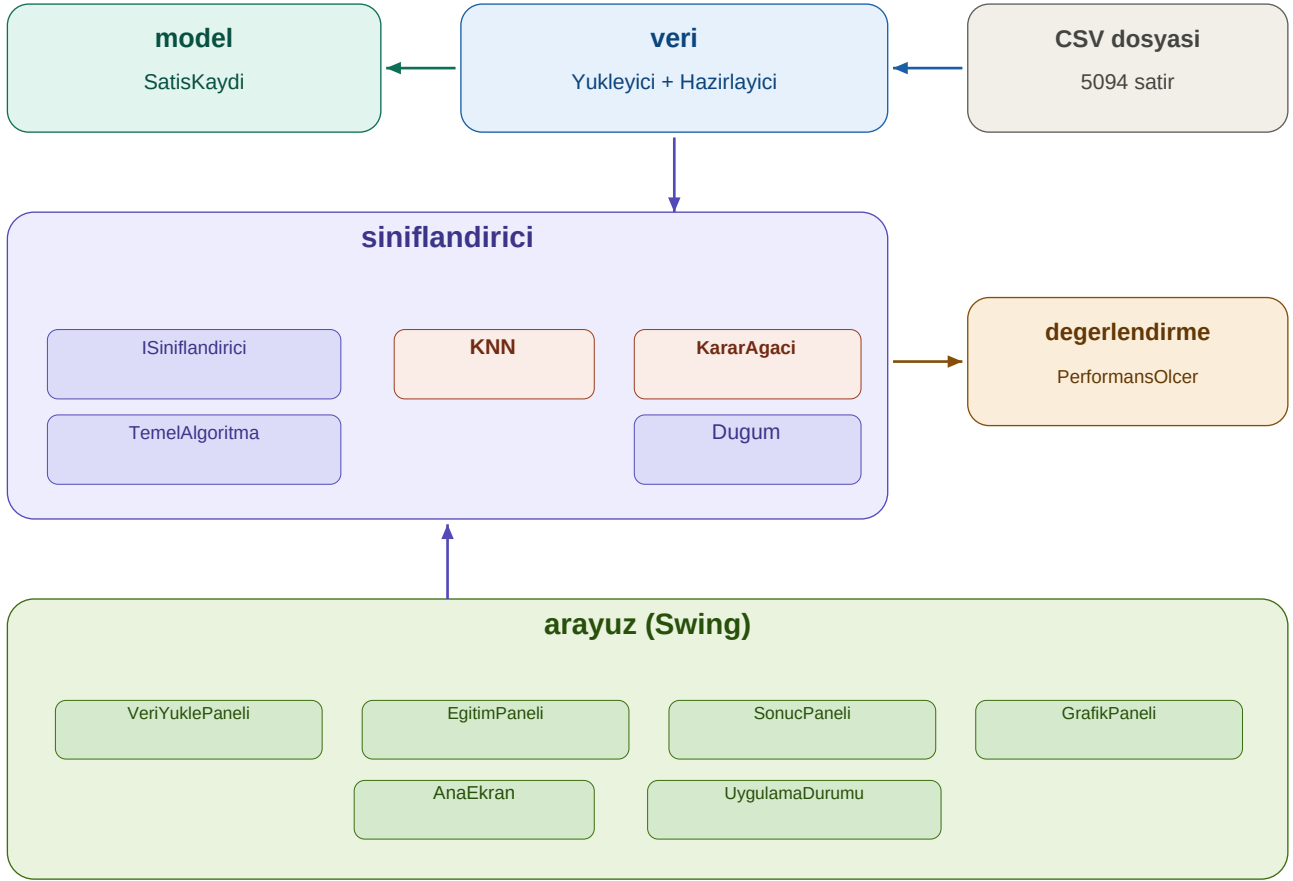
5. Karar Agaci

— Recursive + Info Gain

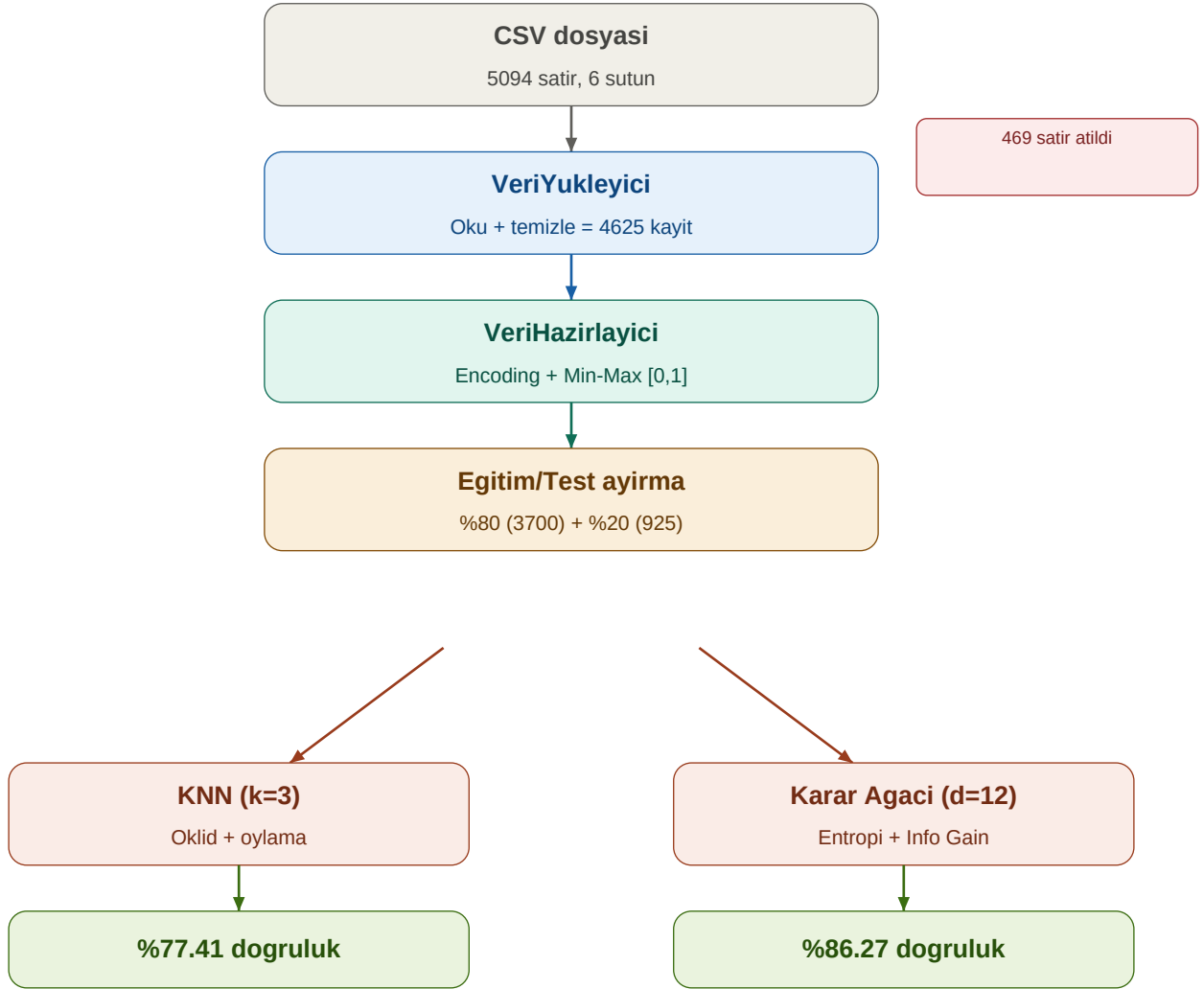
6. Sonuc Karsilastirmasi

— KNN vs DT

1. Genel Mimari — 6 Paket, 12 Sınıf



2. Veri Akis Hatti — CSV'den Sonuca



3. OOP Prensipleri — Kalitim Hiyerarsisi

SOYUTLAMA (Abstraction)

ISiniflandirici (interface)

modeliEgit() | tahminEt() | algoritmaAdi()

Kapsulleme

private+getter

implements

KALITIM (Inheritance)

TemelAlgoritma (abstract)

basariOraniHesapla() | egitimTestAyir()

egitZamanli() — ortak metotlar

extends

COK BICIMLILIK (Polymorphism)

KNNSiniflandirici

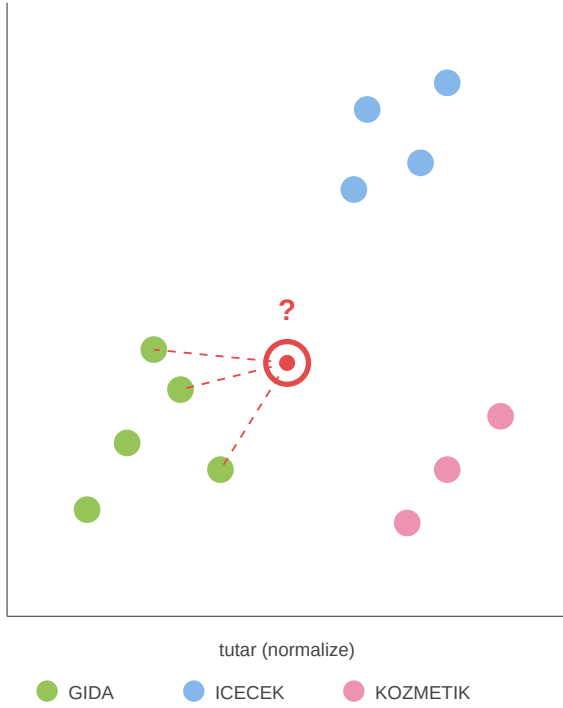
modeliEgit: veriyi sakla
tahminEt: mesafe hesapla

KararAgaciSiniflandirici

modeliEgit: agac kur
tahminEt: agacta gezin

List<TemelAlgoritma> -> ayni metot, farkli davranis

4. KNN Algoritması — Mesafe ve Oylama



Adim 1: Mesafe hesapla

$$d = \sqrt{Dc^2 + Dt^2 + Dm^2}$$

Adim 2: PriorityQueue

Adim 3: Oylama

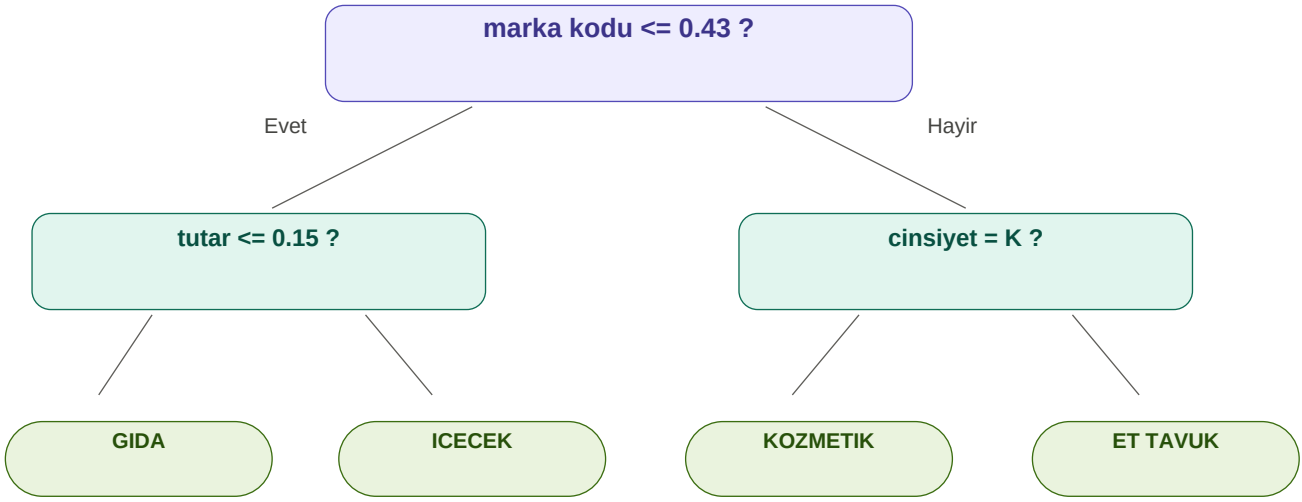
GIDA: 3 oy -> kazandi!

Tahmin: "GIDA"

Performans

Egitim: 0ms | Tahmin: 132ms

5. Karar Agaci — Recursive + Information Gain



Entropi: veri ne kadar karisik?

$$H(S) = -\sum p(i) \times \log_2(p(i))$$

Durdurma kriterleri

maxDepth | minKayit

Bilgi kazanci <= 0

Performans

Egitim: 111ms | Tahmin: 1ms

KNN'den 130x hizli!

6. Sonuc Karsilastirmasi — KNN vs Karar Agaci

